

BVAR vs. Information Criteria: A Monte Carlo Simulation Study

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

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1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

2 Methodology

2.1 Data Generating Process

Explain the true DGP based on Kilian and Murphy (2014) or other literature.

2.2 Optimal Prior Selection via Marginal Data Density

Describe how the shrinkage parameter (τ) is treated as an optimization problem rather than a fixed heuristic.

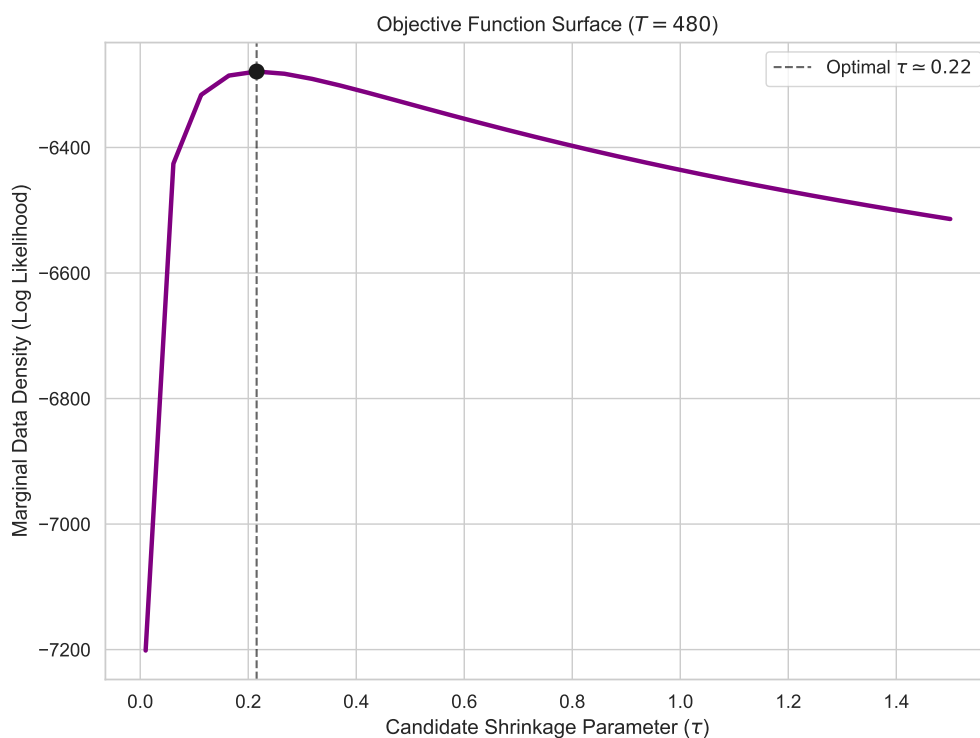


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3 Simulation Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

3.1 Behavior of Standard Information Criteria

Before evaluating the Bayesian estimators, we examine the baseline lag detection performance of the classical Information Criteria.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator	AIC	BIC	HQC
	T			
4	96	12.00	1.50	2.00
	144	2.50	1.00	2.00
	240	3.50	2.00	2.00
	480	3.50	2.00	2.00
6	96	12.00	1.50	2.00
	144	2.50	2.00	2.00
	240	3.00	2.00	2.00
	480	6.00	2.50	2.50
8	96	12.00	1.00	2.00
	144	5.00	1.00	2.00
	240	4.00	1.50	2.00
	480	8.00	2.00	2.00
10	96	12.00	1.00	1.00
	144	9.00	1.50	2.00
	240	2.50	2.00	2.00
	480	9.50	2.00	2.00

3.2 The Uncertainty Strategy: Averaging vs. Selection

Here, we evaluate True Bayesian Model Averaging (BMA) against Information Criteria. We focus on how BMA acts as an insurance policy against catastrophic lag misspecification in small samples.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Model Averaging (Including Informative Priors).

T	p_0	AIC	BIC	HQC	OLS BMA (BIC-W)	OLS BMA (Geom-W, $\theta = 0.5$)	Joint BMA (p, τ)	Geom BMA ($\theta = 0.5$)
96	4	1.403	1.312	1.085	0.689	0.781	-	1.462
	6	0.548	0.926	1.095	0.545	0.861	-	0.251
	8	2.490	1.271	0.801	0.602	0.948	-	0.839
	10	1.262	0.558	0.558	0.235	0.331	-	0.165
144	4	0.883	0.608	0.778	1.409	1.183	1.250	1.238
	6	0.805	0.724	0.724	1.169	0.391	2.900	0.908
	8	2.695	2.067	1.958	1.106	1.968	0.914	0.817
	10	2.706	2.568	2.399	2.550	1.914	3.655	1.628
240	4	1.510	1.100	1.100	2.036	1.157	1.788	1.794
	6	0.310	0.375	0.375	0.639	0.640	0.328	0.671
	8	1.571	1.673	2.481	3.051	0.725	0.923	0.813
	10	3.079	3.362	3.362	2.725	0.770	2.918	0.952
480	4	0.827	1.073	1.073	0.451	0.156	0.463	0.332
	6	1.000	0.613	0.613	2.998	0.169	1.032	1.268
	8	1.000	0.203	0.203	0.434	0.338	0.873	1.445
	10	1.115	1.823	1.823	2.672	0.880	2.109	2.291

Table 3: Posterior Lag-Weight Distribution across all BMA architectures ($p_0 = 4$).

Estimator T Statistic / Lag	OLS-BMA 96	OLS-Geom-BMA 96	Joint-BMA 96	Geom-BMA 96	OLS-BMA 144	OLS-Geom 144
$E[p]$	1.50	1.50	7.69	4.23	1.14	1.08
p=1	50.0%	50.0%	-	0.7%	86.3%	92.4%
p=2	50.0%	50.0%	3.0%	20.7%	13.7%	7.6%
p=3	-	-	10.0%	25.4%	-	-
p=4	-	-	9.4%	13.6%	-	-
p=5	-	-	8.6%	12.9%	-	-
p=6	-	-	8.5%	13.6%	-	-
p=7	-	-	7.6%	6.4%	-	-
p=8	-	-	6.8%	3.2%	-	-
p=9	-	-	8.9%	2.0%	-	-
p=10	-	-	10.1%	0.9%	-	-
p=11	-	-	12.4%	0.6%	-	-
p=12	-	-	14.7%	-	-	-

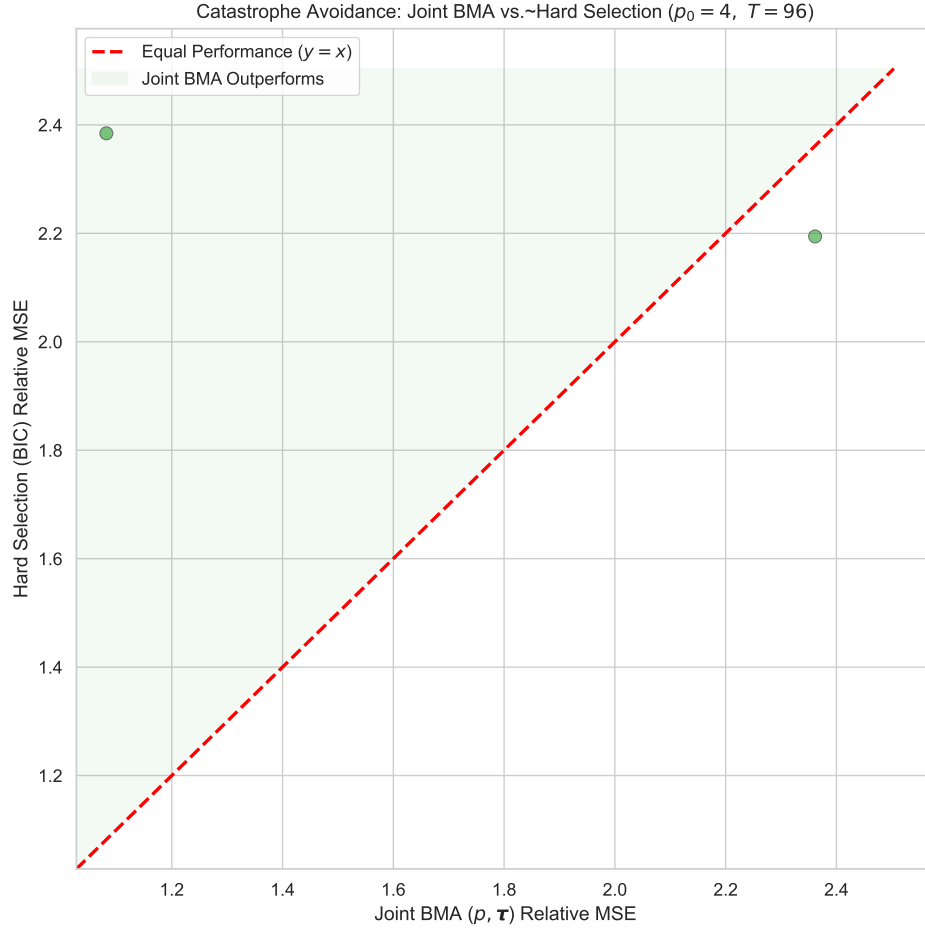


Figure 2: Catastrophe Avoidance: BMA vs. Hard Selection (BIC). Points above the 45-degree line indicate iterations where BMA outperformed BIC, notably smoothing out catastrophic tail risks.

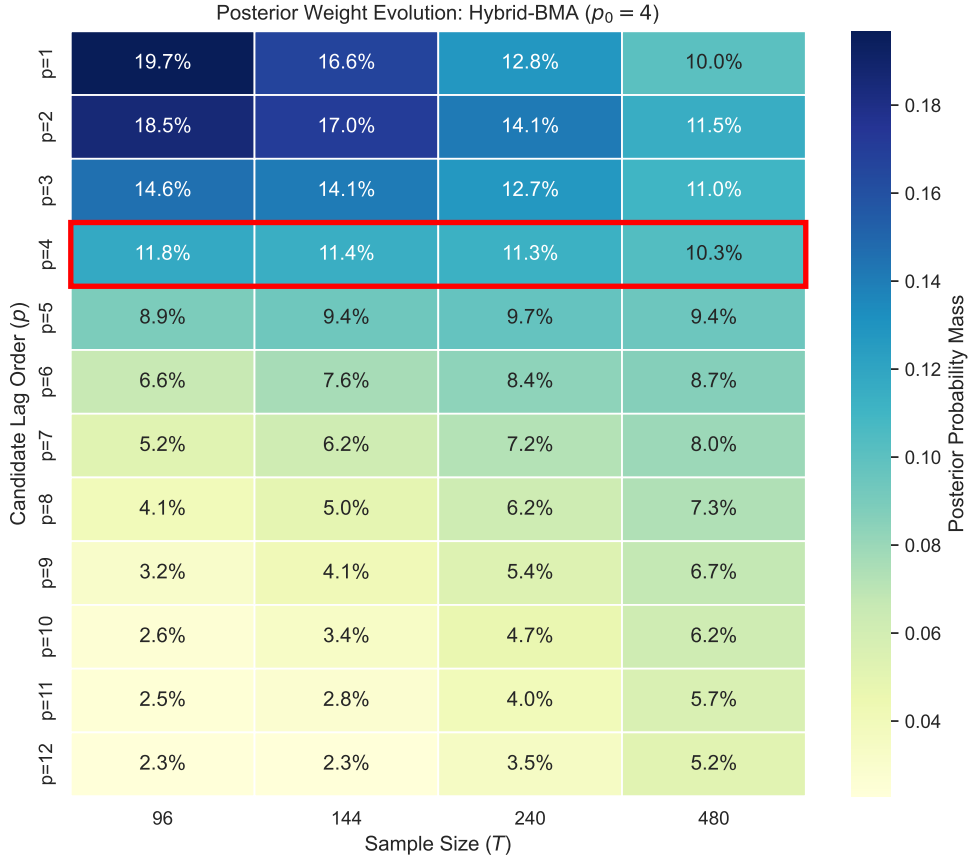


Figure 3: Posterior Weight Evolution of SOTA-BMA. At small sample sizes, probability mass is dispersed to hedge uncertainty. As T grows, the distribution concentrates precisely on the true DGP.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.